# AGENT-STATE: DURABLE CHECKPOINT LOG, VISIBLE BACKUP FAILURES, ONE BAD FILE NO LONGER BLOCKS AN ADAPTER

## 1 INTRODUCTION.

scripts/agent-state.sh copies adapter state from local disk ($AM_LOCAL) to the bucket (/data) every 15 s. From 2026-11-06 the claude adapter failed on every pass for a week; the only trace was a WARN on stdout, and a restart on 2026-11-13 restored a week-old tree. A failure is held to its bad paths, logged durably and shown in the app (figure 1).

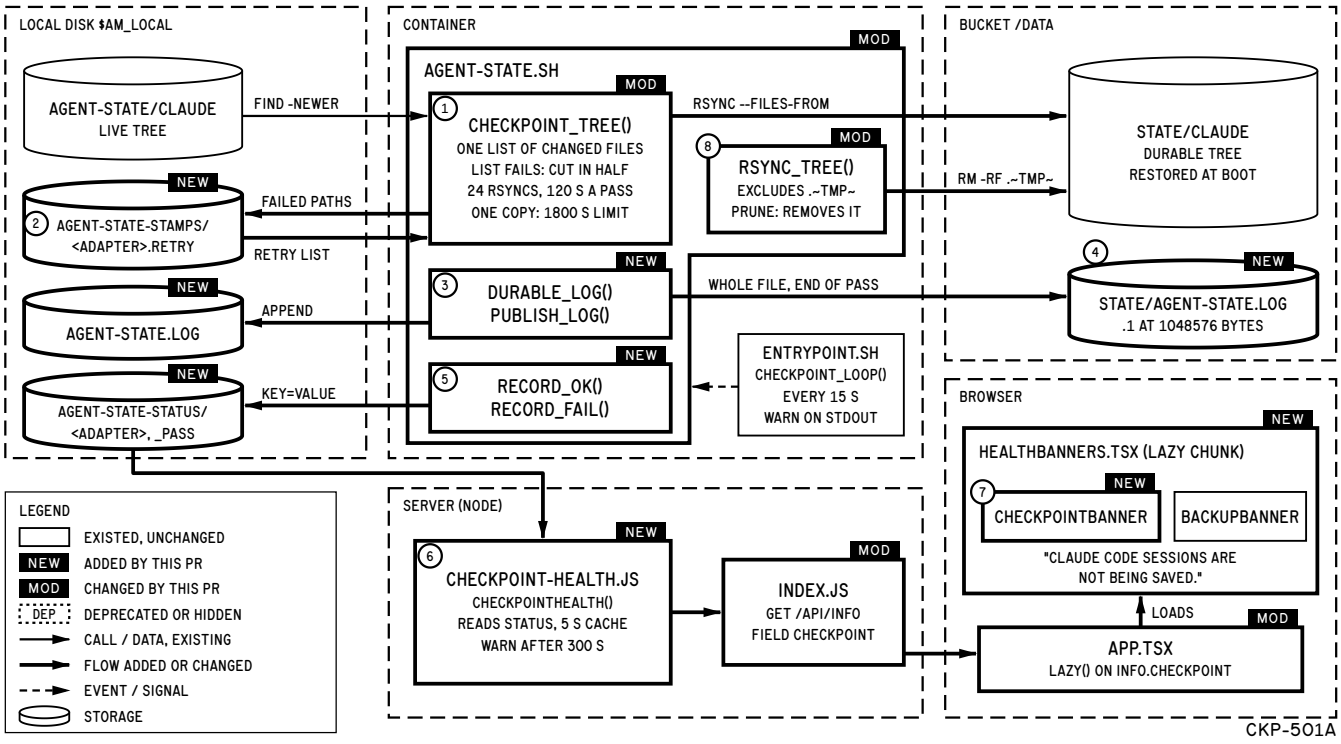



Figure 1. Agent-State Checkpoint, Layout and Data Flow

## 2 HOW IT WORKS.

A list whose rsync fails is cut in half, down to single paths; what copies is durable and the stamp moves on (1). Paths that still fail stay in <adapter>.retry and go as a small list of their own each pass (2). durable_log() writes one line per failure, then every 20th pass while it repeats (3); publish_log() copies the whole log to the bucket, and at 1048576 bytes it rotates to .1 (4). record_ok() and record_fail() write each adapter's outcome (5). checkpointHealth() lists an adapter after 300 s without a clean pass, and /api/info carries it as checkpoint (6). The app then loads the HealthBanners chunk and shows a line that cannot be dismissed (7). .~tmp~ leftovers are no longer copied, and the prune mode removes them (8) (figure 2).

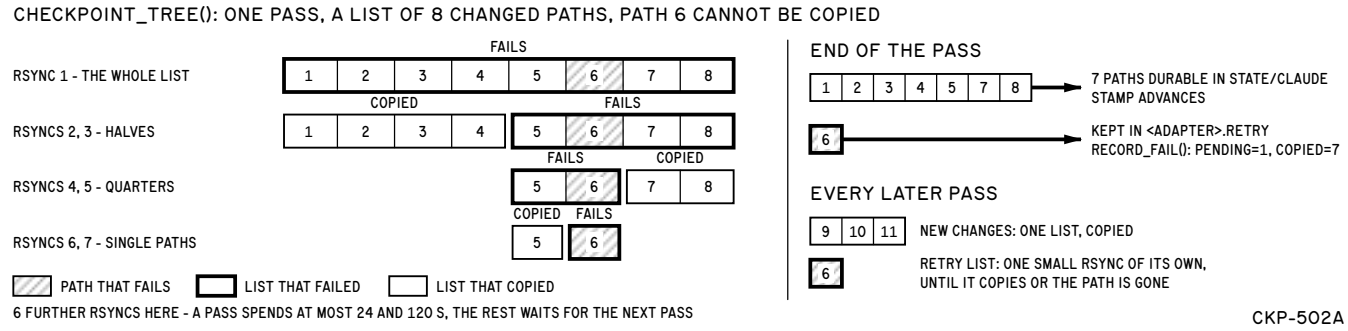



Figure 2. Halving of a Failed List in checkpoint_tree()

## 3 DATA.

Table 1. Checkpoint Limits and Defaults

| Parameter | Value |
|---|---|
| AGENT_STATE_CHECKPOINT_SECONDS (pass interval) | 15 s |
| AGENT_STATE_RETRY_MAX (further rsyncs, per adapter and pass) | 24 |
| AGENT_STATE_RETRY_SECONDS (per adapter and pass) | 120 s |
| AGENT_STATE_CHECKPOINT_TIMEOUT (one copy; 0 = no limit) | 1800 s |
| AGENT_STATE_LOG_MAX (log rotates to .1) | 1048576 bytes |
| AGENT_STATE_LOG_EVERY (unchanged failure logged again) | 20 passes |
| AM_CHECKPOINT_WARN_SECONDS (no clean pass, then warn) | 300 s |

**CAUTION**

Why the passes failed from 2026-11-06 is not known. This PR makes the next failure logged and visible; it does not prove the old cause fixed.

**VERIFY CHECKPOINTS** — **OPERATOR**

```
1 LOG - read the durable log
  √ /data/state/agent-state.log
2 INFO - curl -s localhost:$AM_PORT
    /api/info | jq .checkpoint
  √ null - every adapter is saved
3 DIR - $AM_LOCAL/agent-state-status
  √ failures=0 in each file
4 APP - no warning line shown
```